GEORGIA STATE UNIVERSITY

October 29 · 2026

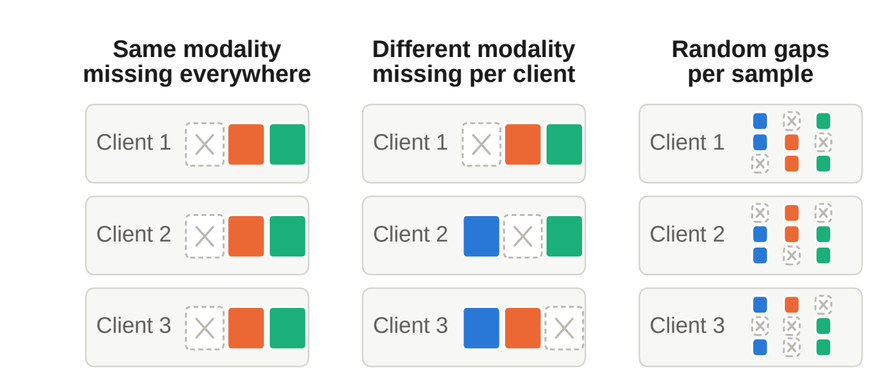
LATENT SYNCHRONIZATION FOR LIGHTWEIGHT MULTIMODAL FEDERATED LEARNING WITH HETEROGENOUS MODALITIES

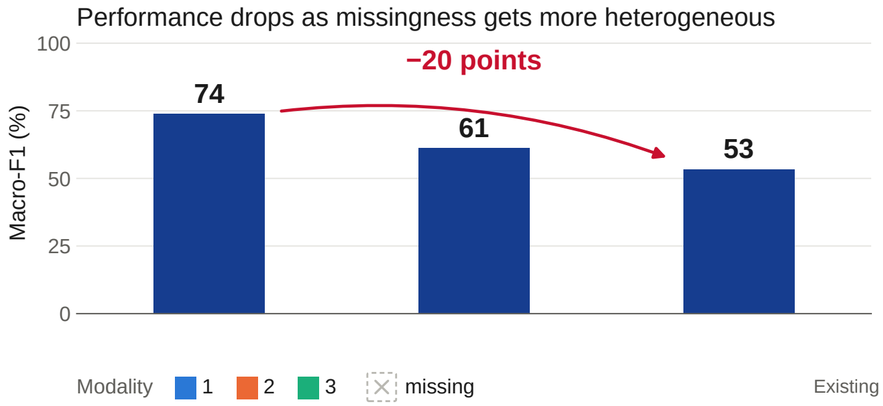
Presenter: Hieu Quach (Jessie)

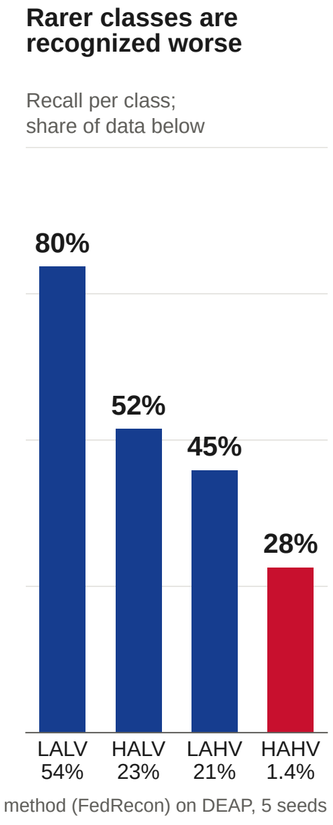
Advisor: Dr. Wei Li (Lisa)

PROBLEM INTRODUCTION

- Robust performance across different scenarios of missing modality under federated learning settings
- Ability to handle multimodal temporal data regardless of domain
- Lightweight model for resource-limited edge clients
- No client owns modality-complete data and missing data must never be used as a training target







RELATED WORK

Multimodal Federated Learning for Missing Modalities

Input-Level

- Idea: a pretrained imputer (an autoencoder) fills in the missing modality before the task model
- Pros: the task model is unchanged and sees a complete input
- Cons: the imputer needs modality-complete training data; it's an extra model

Server-Level

- Idea: cluster clients and replace the parameters most affected by missing data with those of peers in the same cluster
- Pros: nothing changes on the clients
- Cons: it relies on peers that do observe the modality; few works take this approach

Both need someone to have observed the missing modality: complete data for the imputer, or peer clients for the server.

RELATED WORK

Multimodal Federated Learning for Missing Modalities

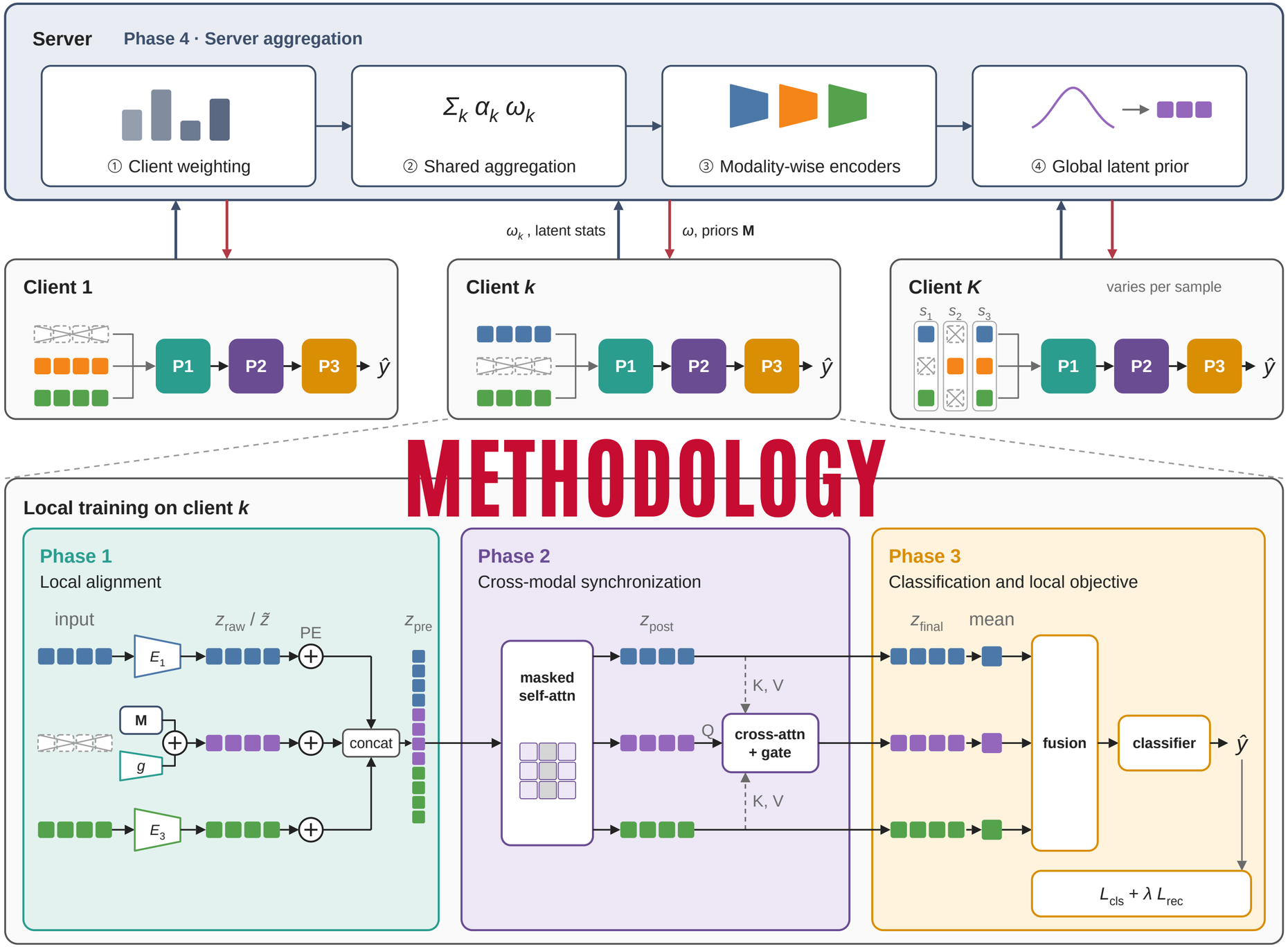
Representation-Level Reconstruction

- Idea: reconstruct missing-modality representations during training
- Pros: one training process; uses information from the other modalities
- Cons: learns from modality-complete pairs or data

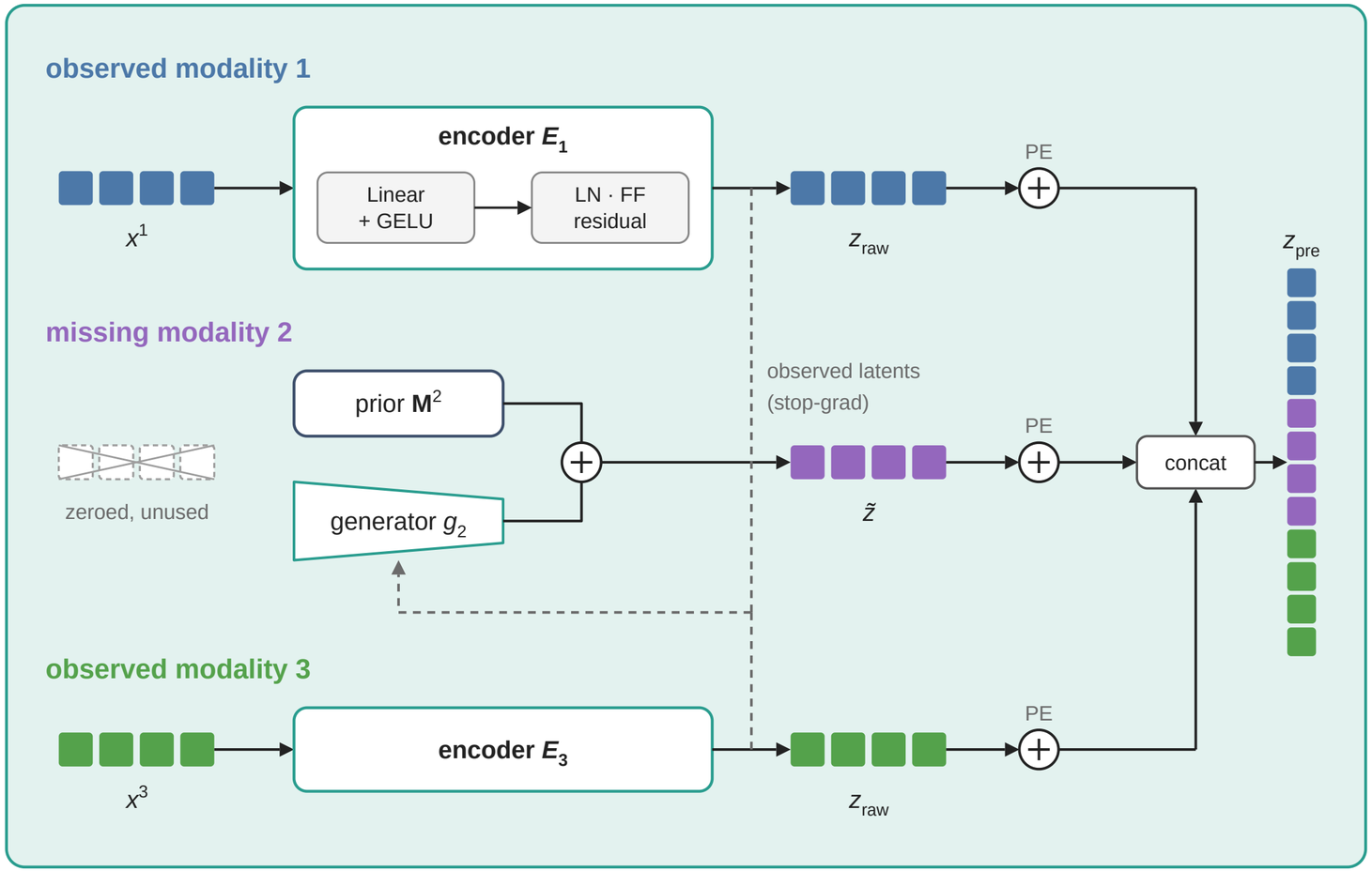
Representation-Level Mitigation

- Idea: keep the gaps and learn representations that are robust to them or consistent across clients
- Pros: no need for reconstruction
- Cons: some rely on modality-complete clients or proxies; often a single missing scenario

Few methods are robust across all missing scenarios; none is tested with a modality absent from every client, and few aim to be lightweight.


Server
Phase 4 · Server aggregation
① Client weighting
② Shared aggregation
③ Modality-wise encoders
④ Global latent prior
ω_k , latent stats
ω, priors **M**
Client 1
Client *k*
Client *K*
varies per sample
P1
P2
P3
$\hat{y}$
METHODOLOGY
Local training on client *k*
Phase 1
Local alignment
input
z_{raw} / $\tilde{z}$
PE
z_{pre}
E_1
E_3
M
g
concat
Phase 2
Cross-modal synchronization
z_{post}
masked self-attn
K, V
Q
cross-attn + gate
Phase 3
Classification and local objective
z_{final}
mean
fusion
classifier
$L_{cls} + \lambda L_{rec}$
observed modality
missing modality
substitute token
client → server
server → clients

PHASE 1: LOCAL ALIGNMENT



Modality Feature Encoder

$$h_\tau^m = \mathrm{GELU}(W^m x_\tau^m + b^m),$$
$$z_{\mathrm{raw},\tau}^m = h_\tau^m + \mathrm{FF}^m\big(\mathrm{LN}(h_\tau^m)\big),$$

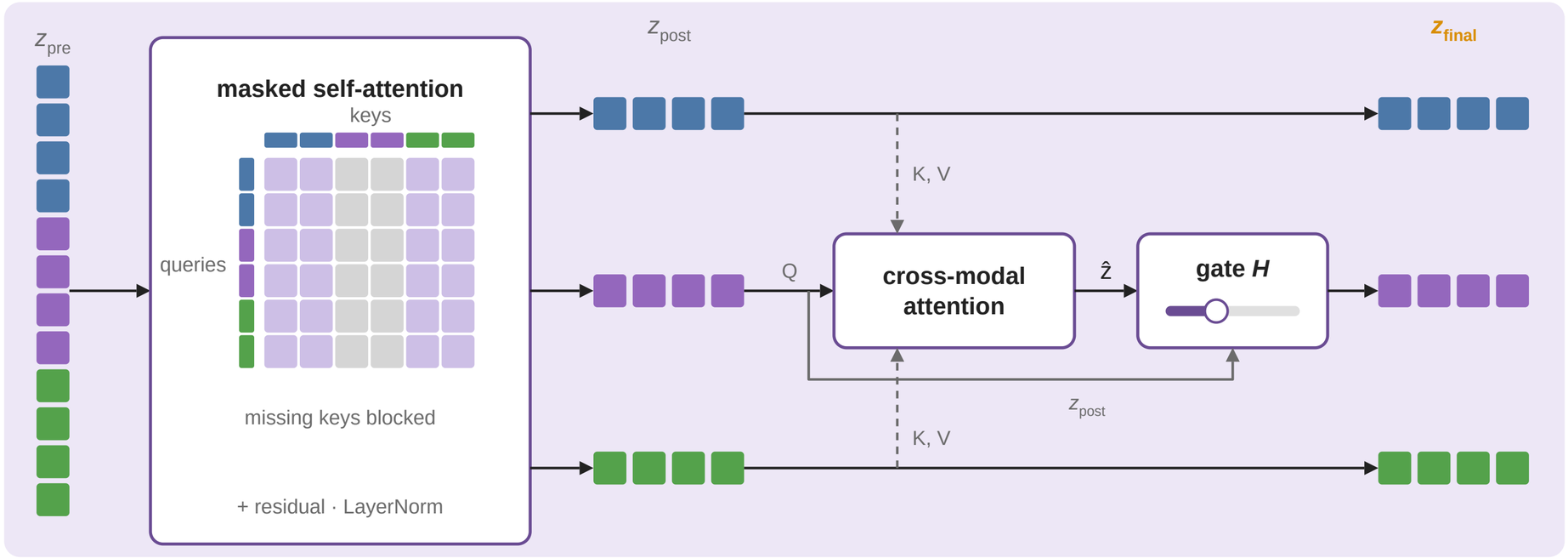
Substitute Tokens

$$\tilde{z}_i^m = \mathbf{M}^m + g_m(u_i,\, a_i),$$
$$u_i = \big[a_i^1\ \mathrm{sg}(z_{\mathrm{raw},i}^1), \ldots, a_i^M\ \mathrm{sg}(z_{\mathrm{raw},i}^M)\big],$$

Positional Encoding

$$z_{\mathrm{pre},i}^m = a_i^m\ z_{\mathrm{raw},i}^m + \big(1 - a_i^m\big)\ \tilde{z}_i^m + \mathrm{PE}.$$

PHASE 2: CROSS-MODAL SYNCHCRONIZATION



Attention Mechanism

$$\mathrm{Attn}(\cdot) = A\mathbf{V}, \quad A = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + B\right),$$

Masked Self-Attention

$$z_{\mathrm{post}} = \mathrm{LN}(z_{\mathrm{pre}} + \mathrm{MHSA}(z_{\mathrm{pre}}; B)),$$

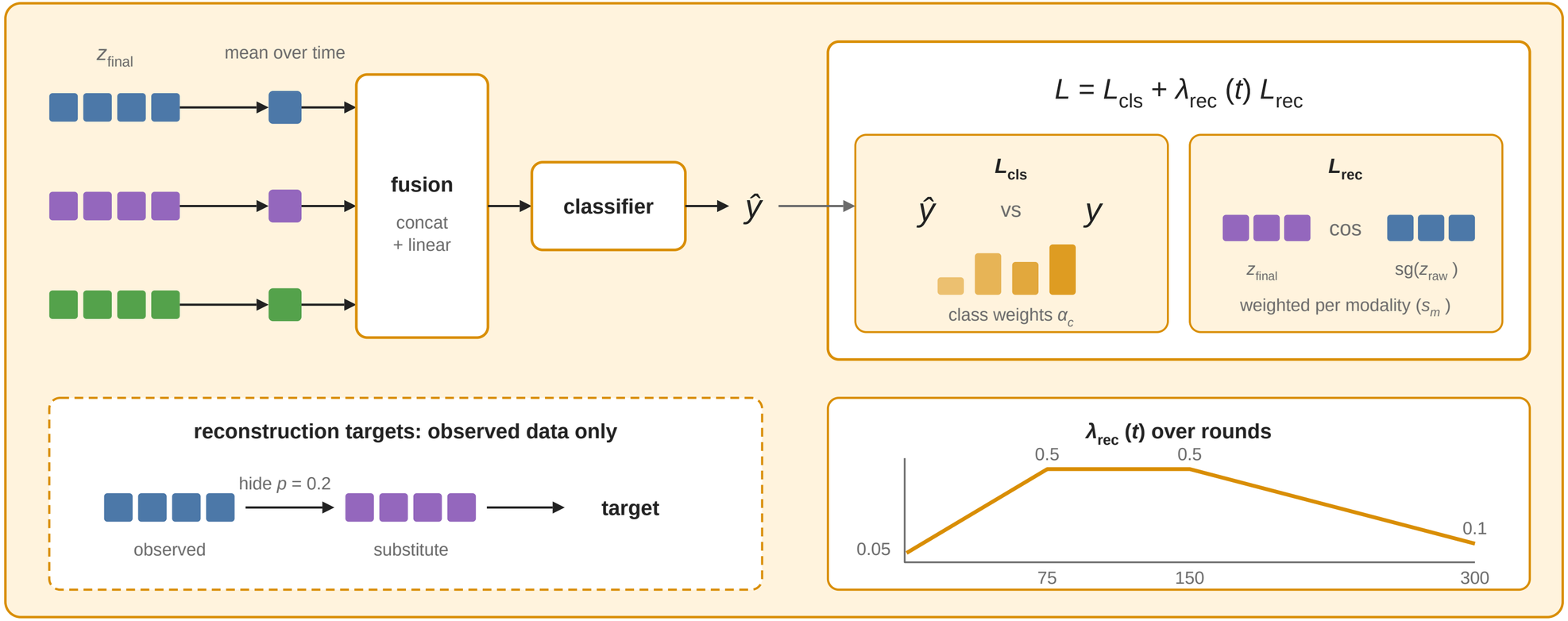
Entropy-Gated Reconstruction

$$\hat{z}^m = \mathrm{LN}(z^m_{\mathrm{post}} + \mathrm{CrossAttn}(z^m_{\mathrm{post}}, \mathcal{C}^m)),$$

$$H^m_\tau = -\frac{1}{\log n_{\mathrm{ctx}}} \sum_{l=1}^{n_{\mathrm{ctx}}} A^m_{\tau l} \log A^m_{\tau l} \in [0,1],$$

$$z^m_{\mathrm{final},\tau} = (1 - H^m_\tau)\, \hat{z}^m_\tau + H^m_\tau\, z^m_{\mathrm{post},\tau},$$

PHASE 3: CLASSIFICATION & OBJECTIVE



Classifier

$$h = \mathrm{GELU}\big(W_f[\bar{z}^1;\ldots;\bar{z}^M] + b_f\big),$$
$$\hat{y} = \mathrm{softmax}\big(W_c\,\mathrm{Dropout}(h) + b_c\big).$$

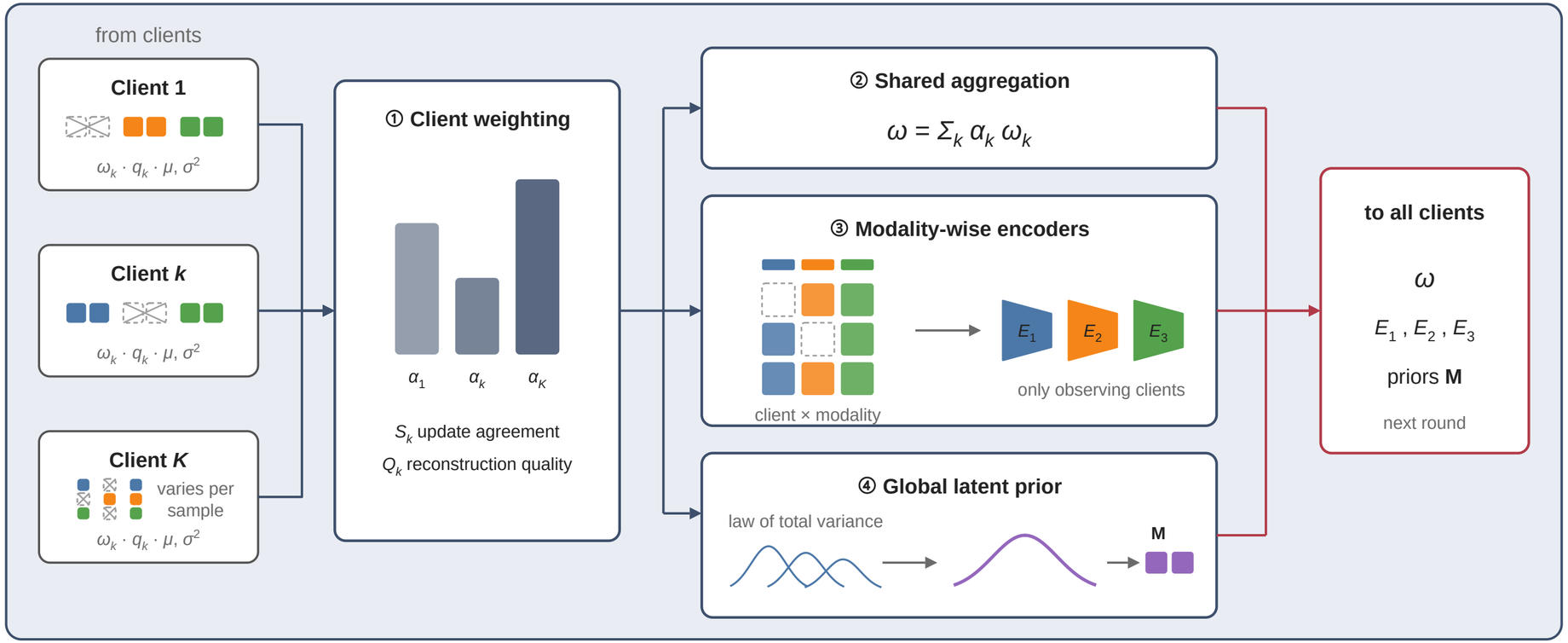
Leak-Free Reconstruction Targets

$$\ell_i^m = \frac{1}{T}\sum_{\tau=1}^{T}\Big(1 - \cos\big(z^m_{\mathrm{final},i,\tau},\ \mathrm{sg}(z^m_{\mathrm{raw},i,\tau})\big)\Big)$$

Leak-Free Reconstruction Targets

$$\mathrm{L_{rec}} = \frac{1}{|\mathcal{A}|}\sum_{m\in\mathcal{A}}\big(e^{-s_m}\,\bar{\ell}^m + s_m\big),$$
$$\mathcal{L} = \mathrm{L_{cls}} + \lambda_{\mathrm{rec}}(t)\,\mathrm{L_{rec}},$$

PHASE 4: GLOBAL AGGREGATION



Similarity Weighting

$$\alpha_k = \frac{n_k \exp\left(\beta(S_k + Q_k)\right)}{\sum_j n_j \exp\left(\beta(S_j + Q_j)\right)},$$

Modality-Wise Encoder Aggregation

$$\omega_E^m = \sum_k \tilde{\alpha}_k^m \, \omega_{E,k}^m, \qquad \tilde{\alpha}_k^m = \frac{\mathbb{1}_k^m \, \alpha_k}{\sum_j \mathbb{1}_j^m \, \alpha_j}$$

Reference Direction

$$\bar{\Delta} \leftarrow \rho \, \bar{\Delta} + (1 - \rho) \sum_k \alpha_k \Delta_k,$$

Global Latent Distribution

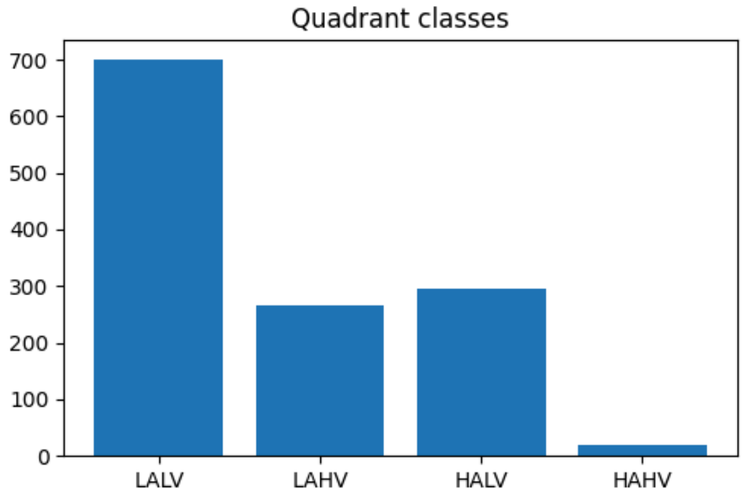
$$\mu^m = \sum_k \tilde{\alpha}_k^m \mu_k^m,$$

$$\sigma^{2,m} = \sum_k \tilde{\alpha}_k^m \sigma_k^{2,m} + \sum_k \tilde{\alpha}_k^m (\mu_k^m - \mu^m)^2,$$

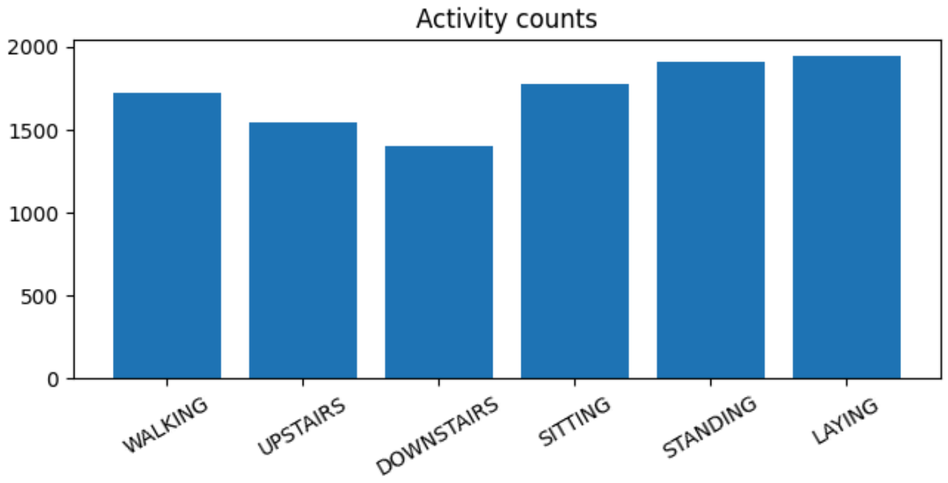
$$\mathbf{M}^m \leftarrow \overline{\mathbf{M}^m} + \gamma^t \left(\mathbf{M}^m - \overline{\mathbf{M}^m}\right),$$

DATASET

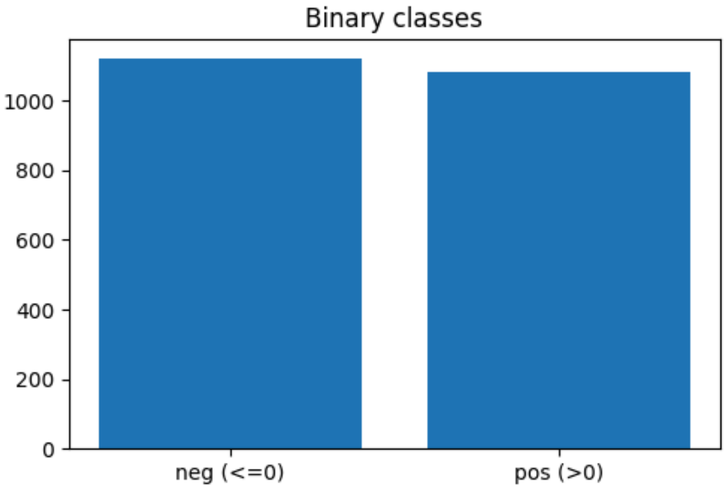
- **DEAP** — 35,840 samples with 3 modalities (EEG, EOG, EMG) and 4 imbalanced classes
- **UCI-HAR** — 10,299 samples with 2 modalities (accelerometer, gyroscope) and 6 nearly balanced classes
- **CMU-MOSI** — 2,199 samples with 3 modalities (acoustic, visual, text) and 2 balanced classes



DEAP



UCI-HAR



CMU-MOSI

MISSINGNESS SIMULATION

There are 20 clients of different sizes, each randomly draws between half and twice the average number of samples (***N LogUniform[0.5N/K, 2N/K]***) from the full dataset which means a sample may go to more than one client.

Missingness is simulated by masking, and missing values are zero-filled for every method:

- **Case 1, absent-*m*:** modality *m* is missing on every client, so it is never observed in the federation
- **Case 1, mixed:** each client misses one randomly chosen modality
- **Case 2:** each modality of each sample is missing with probability $\eta = 0.5$, keeping at least one per sample

EXPERIMENT SETUP

- **Hyperparameters:** 300 rounds · 5 local epochs · Adam, lr = 5×10^{-4} · Batch size 32 · Seeds 42–46 · LaSyM: *d_h = 128;* 8 attention heads; dropout 0.1, encoder/generator hidden 512/32; modality dropout *p = 0.2*; gradient clipping 1.0; *β = 1; ρ = 0.9; γ = 0.995; λ_{rec}: 0.05 → 0.5* (rounds 1–75), held to round 150, → *0.1* at round 300

- **Compared algorithms:** Input-level imputation: FedRecon (arXiv 2025) · Representation-level reconstruction: FedMobile (WWW 2025) · Representation-level robustness: FedInMM (Computer Communications 2024), PEPSY (NeurIPS 2025)

- **Evaluation**: Final-round macro-F1 and accuracy on each client's own test set, averaged over clients (mean ± std over 5 seeds) · Model size = parameters uploaded per client per round

RESULTS

DEAP

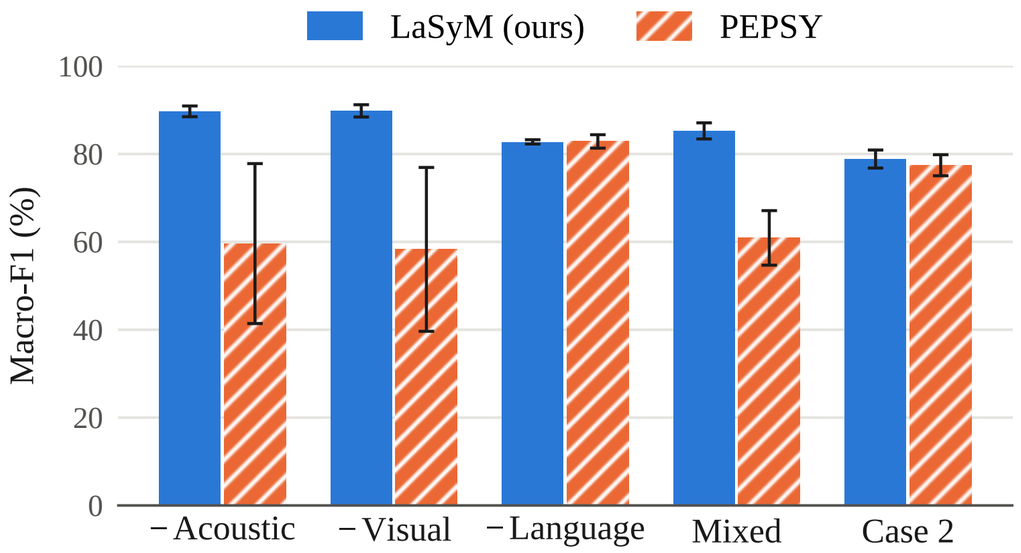
Method	Params	Case 1								Case 2 (η = 0.5)	
		Missing EEG		Missing EOG		Missing EMG		Mixed		Random	
		Acc	*F1*	*Acc*	*F1*	*Acc*	*F1*	*Acc*	*F1*	*Acc*	*F1*
FedInMM	0.69M	78.32±1.42	71.70±1.99	76.73±1.12	69.72±0.68	74.67±1.30	65.67±0.83	64.78±0.90	53.56±0.76	57.75±1.01	42.91±1.71
PEPSY	1.40M	80.80±1.15	75.91±1.32	89.85±1.52	85.03±4.86	89.97±1.70	87.30±1.69	78.66±2.71	71.13±2.99	73.55±1.23	63.55±2.64
FedRecon	4.74M	76.11±1.47	69.36±2.12	81.06±1.23	75.93±1.31	80.62±1.05	75.94±0.87	69.29±2.03	61.10±1.55	65.28±1.15	53.35±1.91
FedMobile	0.66M	54.52±0.36	18.72±1.38	54.60±0.29	18.00±0.40	54.64±0.27	18.60±0.92	53.88±1.37	17.65±0.46	54.61±0.26	20.39±0.45
LaSyM	0.65M	**93.17**±0.79	**91.86**±0.70	**94.89**±0.68	**93.68**±0.91	**94.67**±0.70	**93.31**±0.80	**88.97**±1.26	**86.61**±1.11	**81.50**±1.45	**76.36**±2.02

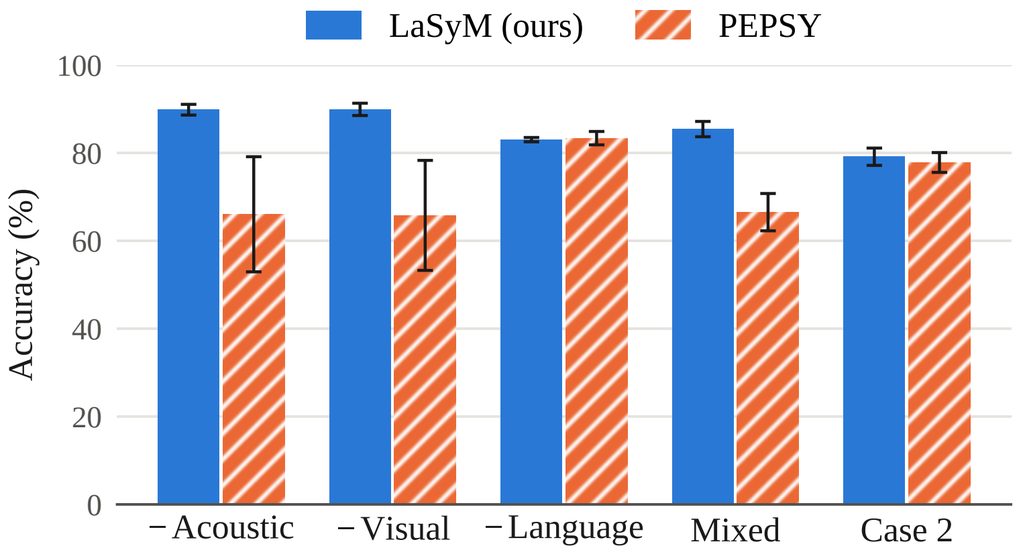
HAR

Method	Params	Case 1						Case 2 (η = 0.5)	
		Missing Acc		Missing Gyro		Mixed		Random	
		Acc	*F1*	*Acc*	*F1*	*Acc*	*F1*	*Acc*	*F1*
FedInMM	0.69M	93.53±1.08	93.87±1.06	97.84±0.42	97.93±0.41	93.01±0.32	93.33±0.30	94.17±0.61	94.38±0.61
PEPSY	1.12M	**96.46**±0.27	**96.64**±0.25	**98.68**±0.40	**98.74**±0.40	91.41±0.54	91.62±0.56	95.29±0.70	95.46±0.67
FedRecon	2.17M	91.24±0.74	91.73±0.68	97.02±0.48	97.17±0.46	92.63±0.41	92.97±0.40	92.09±0.89	92.43±0.83
FedMobile	1.09M	81.17±0.69	81.69±0.66	94.09±0.53	94.21±0.53	23.79±8.39	11.18±8.80	88.32±0.45	88.69±0.46
LaSyM	0.53M	96.07±0.51	96.22±0.50	98.38±0.54	98.44±0.53	**94.84**±0.27	**95.04**±0.28	**96.08**±0.59	**96.24**±0.58

RESULTS

LaSyM vs. PEPSY on CMU-MOSI





THANK YOU